

Output_Logs/Outputs:

User Management

1. Add a member :

Enter Choice: 1

Name: koach

Email: koach@mail.com

Phone: 9191919191

Membership Type Id: 3

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (37ms) [Parameters=[@p0='?' (DbType = DateTime), @p1='?', @p2='?' (DbType = Boolean), @p3='?' (DbType = Int32), @p4='?', @p5='?'], CommandType='Text', CommandTimeout='30']

INSERT INTO "Members" ("CreatedAt", "Email", "IsActive", "MembershipTypeId", "Name", "Phone")

VALUES (@p0, @p1, @p2, @p3, @p4, @p5)

RETURNING "Id";

Member Added

	Id [PK] integer 	Name text 	Email text 	Phone text 	IsActive boolean
1	1	preeta...	preetam@mail.co...	99999999...	true
2	2	ashish	ashish@mail.com	99999000...	true
3	3	vivek	vivek@mail.com	00000999...	true
4	4	koach	koach@mail.com	91919191...	true

2. View all members :

Enter Choice: 2

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (2ms) [Parameters=[], CommandType='Text', CommandTimeout='30']

SELECT m."Id", m."CreatedAt", m."Email", m."IsActive", m."MembershipTypeId", m."Name", m."Phone"

```
FROM "Members" AS m
1 | preetam | preetam@mail.com | 9999999999
2 | ashish | ashish@mail.com | 9999900000
3 | vivek | vivek@mail.com | 0000099999
4 | koach | koach@mail.com | 9191919191
```

3. Search by mail :

Enter Choice: 3

Enter Email: koach@mail.com

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[@__email_0=?'],
CommandType='Text', CommandTimeout='30']
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive",
m."MembershipTypeId", m."Name", m."Phone"
      FROM "Members" AS m
      WHERE m."Email" = @__email_0
      LIMIT 1
```

4 | koach

4. Search By Number :

Enter Choice: 4

Enter Phone Number: 9999999999

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (23ms) [Parameters=[@__phone_0=?'],
CommandType='Text', CommandTimeout='30']
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive",
m."MembershipTypeId", m."Name", m."Phone"
      FROM "Members" AS m
      WHERE m."Phone" = @__phone_0
      LIMIT 1
```

1 | preetam

5. Deactivate a member :

Enter Choice: 5

Member Id: 1

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[@__id_0='?' (DbType =
Int32)], CommandType='Text', CommandTimeout='30']
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive",
m."MembershipTypeId", m."Name", m."Phone", m0."Id",
m0."MaxBorrowDays", m0."MaxBorrowLimit", m0."Name"
      FROM "Members" AS m
      INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId"
= m0."Id"
      WHERE m."Id" = @__id_0
      LIMIT 1
```

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (6ms) [Parameters=[@p6='?' (DbType =
Int32), @p0='?' (DbType = DateTime), @p1='?', @p2='?' (DbType =
Boolean), @p3='?' (DbType = Int32), @p4='?', @p5='?'],
CommandType='Text', CommandTimeout='30']
      UPDATE "Members" SET "CreatedAt" = @p0, "Email" = @p1,
"IsActive" = @p2, "MembershipTypeId" = @p3, "Name" = @p4, "Phone"
= @p5
      WHERE "Id" = @p6;
```

Member Deactivated

```
Enter Choice: 12
Member Id: 1
Book Copy Id: 3
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (3ms) [Parameters=[@__id_0='?' (DbType =
Int32)], CommandType='Text', CommandTimeout='30']
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive", m."
      FROM "Members" AS m
      INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId"
= m0."Id"
      WHERE m."Id" = @__id_0
      LIMIT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[@__p_0='?' (DbType =
Int32)], CommandType='Text', CommandTimeout='30']
      SELECT b."Id", b."BookId", b."Status"
      FROM "BookCopies" AS b
      WHERE b."Id" = @__p_0
      LIMIT 1
Error: member not found or inactive
```

When we try to borrow a book from the deactivated members id, the system should not let us do so.

6. Reactivate a member

Enter Choice: 6

Member Id: 1

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[@__id_0=?' (DbType =
Int32)], CommandType='Text', CommandTimeout='30']
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive",
m."MembershipTypeId", m."Name", m."Phone", m0."Id",
m0."MaxBorrowDays", m0."MaxBorrowLimit", m0."Name"
      FROM "Members" AS m
      INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId"
= m0."Id"
      WHERE m."Id" = @__id_0
      LIMIT 1
```

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (3ms) [Parameters=[@p6=?' (DbType =
Int32), @p0=?' (DbType = DateTime), @p1=?', @p2=?' (DbType =
Boolean), @p3=?' (DbType = Int32), @p4=?', @p5=?'],
CommandType='Text', CommandTimeout='30']
      UPDATE "Members" SET "CreatedAt" = @p0, "Email" = @p1,
"IsActive" = @p2, "MembershipTypeId" = @p3, "Name" = @p4, "Phone"
= @p5
      WHERE "Id" = @p6;
```

Member Reactivated

Lets try borrowing a book from the same account that was reactivated(1)

```

Enter Choice: 12
Member Id: 1
Book Copy Id: 3
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[@__id_0='?' (DbType = Int32), @__id_1='?' (DbType = Int32)], CommandTimeout=30]
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive", m."MembershipTypeId"
      FROM "Members" AS m
      INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId" = m0."Id"
      WHERE m."Id" = @__id_0
      LIMIT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (11ms) [Parameters=[@__id_0='?' (DbType = Int32), @__id_1='?' (DbType = Int32), @__id_2='?' (DbType = Int32)], CommandTimeout=30]
      SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",
      f."BorrowingId", f."IsPaid", f."PaidDate"
      FROM "Borrowings" AS b
      INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"
      LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"
      WHERE b."ReturnDate" IS NULL AND b."MemberId" = @__id_0
Error: borrowing limit reached for this member

```

We get this when a member is active and has reached borrowing limit(which is 2 for basic membership)

Book Management

7. Add Book

```

Enter Choice: 7
Title: Metamorphosis
Author: Franz Kafka
Category Id: 3
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (7ms) [Parameters=[@p0='?', @p1='?' (DbType = Int32), @p2='?'], CommandType='Text', CommandTimeout=30]
      INSERT INTO "Books" ("Author", "CategoryId", "Title")
      VALUES (@p0, @p1, @p2)
      RETURNING "Id";
Book Added

```

8. Add Multiple Copies

Enter Choice: 10

Book Id: 4

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (4ms) [Parameters=[@p0='?' (DbType = Int32), @p1='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']
      INSERT INTO "BookCopies" ("BookId", "Status")
      VALUES (@p0, @p1)
      RETURNING "Id";
```

Book Copy Added

	Id [PK] integer 	Status integer 	BookId integer 
5	6	1	2
6	3	1	1
7	4	0	1
8	8	0	3
9	9	1	3
10	10	1	4
11	11	1	4
12	12	1	4

3 new copies were added(manually done thrice) with status being 1 - Available

9. View Books(Available)

Enter Choice: 8

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (5ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT b."Id", b."Author", b."CategoryId", b."Title"
      FROM "Books" AS b
```

- 1 | clean code | robert martin
- 2 | System Design | Alex Xu
- 3 | Operating Systems Concepts | Abraham
- 4 | Metamorphosis | Franz Kafka

10. Search By title

Enter Choice: 10

Enter Title: Design

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (2ms)

[Parameters=[@__title_0_contains=?], CommandType='Text',
CommandTimeout='30']

SELECT b."Id", b."Author", b."CategoryId", b."Title"

FROM "Books" AS b

WHERE b."Title" LIKE @__title_0_contains

2 | System Design

Enter Choice: 10

Enter Title: clean

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (12ms)

[Parameters=[@__title_0_contains=?], CommandType='Text',
CommandTimeout='30']

SELECT b."Id", b."Author", b."CategoryId", b."Title"

FROM "Books" AS b

WHERE b."Title" LIKE @__title_0_contains

1 | clean code

11. Search by Author

Enter Choice: 11

Enter Author's Name: Franz

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (23ms)

[Parameters=[@__author_0_contains=?], CommandType='Text',
CommandTimeout='30']

SELECT b."Id", b."Author", b."CategoryId", b."Title"

FROM "Books" AS b

WHERE b."Author" LIKE @__author_0_contains
4 | Metamorphosis

12. Mark a copy as damaged

Enter Choice: 12

Book Copy Id: 9

info: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (3ms) [Parameters=[@__bookCopyId_0='?'
(DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT b."Id", b."BookId", b."Status"
```

```
FROM "BookCopies" AS b
```

```
WHERE b."Id" = @__bookCopyId_0
```

```
LIMIT 1
```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (2ms) [Parameters=[@p1='?' (DbType =
Int32), @p0='?' (DbType = Int32)], CommandType='Text',
CommandTimeout='30']

```
UPDATE "BookCopies" SET "Status" = @p0
```

```
WHERE "Id" = @p1;
```

Book Copy Marked Damaged

	Id [PK] integer 	Status integer 	BookId integer 
5	3	1	1
6	4	0	1
7	8	0	3
8	10	1	4
9	11	1	4
10	12	1	4
11	1	2	1
12	9	2	3

Borrowing rules:

13. Testing Basic Membership

Enter Choice: 13

Member Id: 5

Book Copy Id: 9

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (2ms) [Parameters=[@__id_0=?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT m."Id", m."CreatedAt", m."Email", m."IsActive",  
m."MembershipTypeId", m."Name", m."Phone", m0."Id",  
m0."MaxBorrowDays", m0."MaxBorrowLimit", m0."Name"  
FROM "Members" AS m  
INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId"  
= m0."Id"  
WHERE m."Id" = @__id_0  
LIMIT 1
```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (1ms) [Parameters=[@__p_0=?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT b."Id", b."BookId", b."Status"  
FROM "BookCopies" AS b  
WHERE b."Id" = @__p_0  
LIMIT 1
```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (1ms) [Parameters=[@__id_0=?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",  
b."MemberId", b."ReturnDate", b."Status", b0."Id", b0."BookId",  
b0."Status", f."Id", f."Amount", f."BorrowingId", f."IsPaid", f."PaidDate"  
FROM "Borrowings" AS b  
INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"  
LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"  
WHERE b."ReturnDate" IS NULL AND b."MemberId" = @__id_0
```

**Error: borrowing limit reached for this member :— Student
Membership - already borrowed 3 copies**

14. Return Service :

Lets return a bookCopy from user 5

Enter Choice: 14

Borrowing Id: 10 — MemberId = 5, BookCopyId = 7

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (6ms) [Parameters=[@__id_0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",  
b."MemberId", b."ReturnDate", b."Status", b0."Id", b0."BookId",  
b0."Status", f."Id", f."Amount", f."BorrowingId", f."IsPaid", f."PaidDate"  
FROM "Borrowings" AS b  
INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"  
LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"  
WHERE b."Id" = @__id_0  
LIMIT 1
```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (2ms) [Parameters=[@p6='?' (DbType = Int32), @p0='?' (DbType = Int32), @p1='?' (DbType = DateTime), @p2='?' (DbType = DateTime), @p3='?' (DbType = Int32), @p4='?' (DbType = DateTime), @p5='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
UPDATE "Borrowings" SET "BookCopyId" = @p0, "BorrowDate" =  
@p1, "DueDate" = @p2, "MemberId" = @p3, "ReturnDate" = @p4,  
"Status" = @p5  
WHERE "Id" = @p6;
```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (1ms) [Parameters=[@p2='?' (DbType = Int32), @p0='?' (DbType = Int32), @p1='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
UPDATE "BookCopies" SET "BookId" = @p0, "Status" = @p1  
WHERE "Id" = @p2;
```

Book Returned

14. Duplicate Borrow :

User 5 had already borrowed book copy with bookId = 4, bookCopyId = 10

Lets try borrowing another with same bookId say bookCopyId = 12

```

Enter Choice: 13
Member Id: 5
Book Copy Id: 12
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[@__id_0='?' (DbType = Int32)], CommandText='
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive", m."MembershipTypeId", m."Name", m."Phone", m0."Id",
      m0."MaxBorrowDays", m0."MaxBorrowLimit", m0."Name"
      FROM "Members" AS m
      INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId" = m0."Id"
      WHERE m."Id" = @__id_0
      LIMIT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (1ms) [Parameters=[@__id_0='?' (DbType = Int32)], CommandText='
      SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate", b."ReturnDate",
      f."BorrowingId", f."IsPaid", f."PaidDate"
      FROM "Borrowings" AS b
      INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"
      LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"
      WHERE b."ReturnDate" IS NULL AND b."MemberId" = @__id_0
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[p0='?' (DbType = Int32)], CommandText='
      SELECT calculate_member_fine(@p0)
Error: book already borrowed

```

15. Unavailable Copy

```

Enter Choice: 13
Member Id: 5
Book Copy Id: 10
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (4ms) [Parameters=[@__id_0='?' (DbType = Int32)], CommandText='
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive", m."MembershipTypeId", m."Name", m."Phone", m0."Id",
      m0."MaxBorrowDays", m0."MaxBorrowLimit", m0."Name"
      FROM "Members" AS m
      INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId" = m0."Id"
      WHERE m."Id" = @__id_0
      LIMIT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]

```

Executed DbCommand (3ms) [Parameters=[@__id_0=?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",  
b."MemberId", b."ReturnDate", b."Status", b0."Id", b0."BookId",  
b0."Status", f."Id", f."Amount", f."BorrowingId", f."IsPaid", f."PaidDate"  
FROM "Borrowings" AS b  
INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"  
LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"  
WHERE b."ReturnDate" IS NULL AND b."MemberId" = @__id_0
```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (10ms) [Parameters=[p0=?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT calculate_member_fine(@p0)
```

Error: book is not found or not available

16. Fine Limit Block

```
INSERT INTO "Fines"  
("Amount", "IsPaid", "BorrowingId")  
VALUES  
(600, false, 1);
```

A fine is manually inserted into the db for user 1
Lets try borrowing for that user

```

Enter Choice: 13
Member Id: 1
Book Copy Id: 6
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[@__id_0='?' (DbType = Int)]
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive", m."
      FROM "Members" AS m
      INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId" = m0."Id"
      WHERE m."Id" = @__id_0
      LIMIT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (1ms) [Parameters=[@__id_0='?' (DbType = Int)]
      SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",
      f."BorrowingId", f."IsPaid", f."PaidDate"
      FROM "Borrowings" AS b
      INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"
      LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"
      WHERE b."ReturnDate" IS NULL AND b."MemberId" = @__id_0
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (1ms) [Parameters=[p0='?' (DbType = Int)]
      SELECT calculate_member_fine(@p0)
Error: pay fine first

```

It is required to pay the fine first(>500) in order to borrow another book

17. OverDue

Should Generate Fines accordingly

- - due date pushed back 4 days - should generate 40rs fine

UPDATE "Borrowings"

SET "DueDate" = NOW() - INTERVAL '4 days'

WHERE "Id" = 17;

Fine is generated after the book is returned

Enter Choice: 14

Borrowing Id: 17

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (6ms) [Parameters=[@__id_0='?' (DbType = Int)]

```

Int32)], CommandType='Text', CommandTimeout='30']
    SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",
    b."MemberId", b."ReturnDate", b."Status", b0."Id", b0."BookId",
    b0."Status", f."Id", f."Amount", f."BorrowingId", f."IsPaid", f."PaidDate"
    FROM "Borrowings" AS b
    INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"
    LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"
    WHERE b."Id" = @__id_0
    LIMIT 1

```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]
 Executed DbCommand (1ms) [Parameters=[@p6='?' (DbType = Int32), @p0='?' (DbType = Int32), @p1='?' (DbType = DateTime), @p2='?' (DbType = DateTime), @p3='?' (DbType = Int32), @p4='?' (DbType = DateTime), @p5='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```

    UPDATE "Borrowings" SET "BookCopyId" = @p0, "BorrowDate" =
    @p1, "DueDate" = @p2, "MemberId" = @p3, "ReturnDate" = @p4,
    "Status" = @p5
    WHERE "Id" = @p6;

```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]
 Executed DbCommand (0ms) [Parameters=[@p2='?' (DbType = Int32), @p0='?' (DbType = Int32), @p1='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```

    UPDATE "BookCopies" SET "BookId" = @p0, "Status" = @p1
    WHERE "Id" = @p2;

```

Fine added

Book Returned

	Id [PK] integer 	Amount numeric (10,2) 	IsPaid boolean 	PaidDate timestamp with time zo
1	1	0.00	false	[null]
2	2	0.00	false	[null]
3	3	50.00	false	[null]

18. Transaction Rollback

Added a throw new Exception("rollback testing"); // to simulate a real-world scenario where any of the previous exceptions might occur during execution.
Right before transaction.commit();

Enter Choice: 13

Member Id: 6

Book Copy Id: 18

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (23ms) [Parameters=[@__id_0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive",
m."MembershipTypeId", m."Name", m."Phone", m0."Id",
m0."MaxBorrowDays", m0."MaxBorrowLimit", m0."Name"
      FROM "Members" AS m
      INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId"
= m0."Id"
      WHERE m."Id" = @__id_0
      LIMIT 1
```

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[@__p_0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']
      SELECT b."Id", b."BookId", b."Status"
      FROM "BookCopies" AS b
      WHERE b."Id" = @__p_0
      LIMIT 1
```

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (3ms) [Parameters=[@__id_0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']
      SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",
b."MemberId", b."ReturnDate", b."Status", b0."Id", b0."BookId",
b0."Status", f."Id", f."Amount", f."BorrowingId", f."IsPaid", f."PaidDate"
      FROM "Borrowings" AS b
      INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"
      LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"
      WHERE b."ReturnDate" IS NULL AND b."MemberId" = @__id_0
```

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (4ms) [Parameters=[p0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']
      SELECT calculate_member_fine(@p0)
```

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (2ms) [Parameters=[@p0='?' (DbType =
```

```

Int32), @p1='?' (DbType = DateTime), @p2='?' (DbType = DateTime),
@p3='?' (DbType = Int32), @p4='?' (DbType = DateTime), @p5='?'
(DbType = Int32)], CommandType='Text', CommandTimeout='30']
    INSERT INTO "Borrowings" ("BookCopyId", "BorrowDate",
"DueDate", "MemberId", "ReturnDate", "Status")
    VALUES (@p0, @p1, @p2, @p3, @p4, @p5)
    RETURNING "Id";

```

```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]
    Executed DbCommand (1ms) [Parameters=[@p2='?' (DbType =
Int32), @p0='?' (DbType = Int32), @p1='?' (DbType = Int32)],
CommandType='Text', CommandTimeout='30']
    UPDATE "BookCopies" SET "BookId" = @p0, "Status" = @p1
    WHERE "Id" = @p2;

```

Error: rollback test

We'll verify by checking if any new borrowings have been added
 These are the last transactions :

17	17	2026-05-20 00:32:48.362349+05:...	2026-05-16 00:33:22
18	18	2026-05-20 00:37:33.981111+05:...	2026-05-27 00:37:33

Enter Choice: 13

Member Id: 2

Book Copy Id: 18

This is what we used as test sample and that failed to reflect in the
 borrowings table proving rollbacks worked

Reports

19. Currently Borrowed books

Enter Choice: 16

```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]
    Executed DbCommand (3ms) [Parameters=[], CommandType='Text',
CommandTimeout='30']
    SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",
b."MemberId", b."ReturnDate", b."Status"
    FROM "Borrowings" AS b
    WHERE b."ReturnDate" IS NULL

```


Borrowing Id: 8 | Member Id: 3
 Borrowing Id: 9 | Member Id: 5
 Borrowing Id: 14 | Member Id: 5

Id [PK] integer 	BorrowDate timestamp with time zone 	DueDate timestamp with time zone
1	2026-05-19 14:19:41.541557+05:...	2026-05-14 14:28:42.33463
3	2026-05-19 14:27:53.727453+05:...	2026-05-26 14:27:53.72745
5	2026-05-19 15:09:36.67653+05:30	2026-05-26 15:09:36.67655
4	2026-05-19 14:29:51.049765+05:...	2026-05-14 15:17:53.98063
8	2026-05-19 16:48:13.162689+05:...	2026-05-29 16:48:13.16272
9	2026-05-19 23:49:54.883421+05:...	2026-05-29 23:49:54.88343
12	2026-05-19 23:59:22.913935+05:...	2026-05-29 23:59:22.91393
6	2026-05-19 16:07:59.396648+05:...	2026-05-26 16:07:59.39666
7	2026-05-19 16:12:55.446186+05:...	2026-05-26 16:12:55.44619
13	2026-05-20 00:06:36.019393+05:...	2026-05-16 00:15:24.35828
10	2026-05-19 23:52:34.649443+05:...	2026-05-16 00:18:32.27413
14	2026-05-20 00:22:18.969617+05:...	2026-05-30 00:22:18.96967

Id here is borrowing Id(from Borrowings table) and 0 indicates that books are currently borrowed 1 indicates they're returned

20. Borrow history by member

```

Enter Choice: 17
Member Id: 5
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (3ms) [Parameters=[@__id_0='?' (DbType=
      SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate"
      FROM "Borrowings" AS b
      WHERE b."MemberId" = @__id_0
      Borrowing Id: 9 | Borrow Date: 19/05/2026 6:19:54 PM
      Borrowing Id: 10 | Borrow Date: 19/05/2026 6:22:34 PM
      Borrowing Id: 11 | Borrow Date: 19/05/2026 6:22:52 PM
      Borrowing Id: 12 | Borrow Date: 19/05/2026 6:29:22 PM
      Borrowing Id: 13 | Borrow Date: 19/05/2026 6:36:36 PM
      Borrowing Id: 14 | Borrow Date: 19/05/2026 6:52:18 PM

```

21. Get Active Borrowings

```

Enter Choice: 16
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (1ms) [Parameters=[], CommandType='Text
      SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate"
      FROM "Borrowings" AS b
      WHERE b."ReturnDate" IS NULL
      Borrowing Id: 8 | Member Id: 3
      Borrowing Id: 9 | Member Id: 5
      Borrowing Id: 14 | Member Id: 5

```

22. Get Pending Fines By MemberId

```

Enter Choice: 15
Member Id: 6
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (44ms) [Parameters=[@__id_0='?' (DbType=
      SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",
      idDate", b0."Id", b0."BookId", b0."Status", m."Id", m."CreatedAt"
      FROM "Borrowings" AS b
      LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"
      INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"
      INNER JOIN "Members" AS m ON b."MemberId" = m."Id"
      WHERE b."MemberId" = @__id_0
      Fine Id: 3 | Amount: 50.00

```

23. Most Borrowed Books

```

Enter Choice: 23
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (13ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",
      b."CategoryId", b1."Title"
      FROM "Borrowings" AS b
      INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"
      INNER JOIN "Books" AS b1 ON b0."BookId" = b1."Id"
Title: clean code | Borrow Count: 10
Title: System Design | Borrow Count: 4
Title: Metamorphosis | Borrow Count: 2
Title: Operating Systems Concepts | Borrow Count: 1
Title: Computer Networks | Borrow Count: 1

```

Validation Tests :

24. Validating members

```

Enter Choice: 13
Member Id: 100
Book Copy Id: 212
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (26ms) [Parameters=[@__id_0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']
      SELECT m."Id", m."CreatedAt", m."Email", m."IsActive",
      m."MembershipTypeId", m."Name", m."Phone", m0."Id",
      m0."MaxBorrowDays", m0."MaxBorrowLimit", m0."Name"
      FROM "Members" AS m
      INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId"
      = m0."Id"
      WHERE m."Id" = @__id_0
      LIMIT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (3ms) [Parameters=[@__p_0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']
      SELECT b."Id", b."BookId", b."Status"
      FROM "BookCopies" AS b
      WHERE b."Id" = @__p_0
      LIMIT 1

```

Error: member not found or inactive

25. Validating Invalid book copies

Enter Choice: 13

Member Id: 3

Book Copy Id: 100

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (16ms) [Parameters=[@__id_0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT m."Id", m."CreatedAt", m."Email", m."IsActive",  
m."MembershipTypeId", m."Name", m."Phone", m0."Id",  
m0."MaxBorrowDays", m0."MaxBorrowLimit", m0."Name"  
FROM "Members" AS m  
INNER JOIN "MembershipTypes" AS m0 ON m."MembershipTypeId"  
= m0."Id"  
WHERE m."Id" = @__id_0  
LIMIT 1
```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (2ms) [Parameters=[@__p_0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT b."Id", b."BookId", b."Status"  
FROM "BookCopies" AS b  
WHERE b."Id" = @__p_0  
LIMIT 1
```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (8ms) [Parameters=[@__id_0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT b."Id", b."BookCopyId", b."BorrowDate", b."DueDate",  
b."MemberId", b."ReturnDate", b."Status", b0."Id", b0."BookId",  
b0."Status", f."Id", f."Amount", f."BorrowingId", f."IsPaid", f."PaidDate"  
FROM "Borrowings" AS b  
INNER JOIN "BookCopies" AS b0 ON b."BookCopyId" = b0."Id"  
LEFT JOIN "Fines" AS f ON b."Id" = f."BorrowingId"  
WHERE b."ReturnDate" IS NULL AND b."MemberId" = @__id_0
```

info: Microsoft.EntityFrameworkCore.Database.Command[20101]

Executed DbCommand (48ms) [Parameters=[p0='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']

```
SELECT calculate_member_fine(@p0)
```

Error: book is not found or not available

